## PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

## ARTICLE DETAILS

| | |
|---|---|
| **TITLE (PROVISIONAL)** | How is television time linked to cardiometabolic health in adults? A critical systematic review of the evidence for an effect of watching television on eating, movement, affect and sleep |
| **AUTHORS** | Wagnild, Janelle; Pollard, Tessa |

## VERSION 1 – REVIEW

| | |
|---|---|
| **REVIEWER** | Sarah Keadle<br>California Polytechnic State University San Luis Obispo |
| **REVIEW RETURNED** | 02-Dec-2020 |

| | |
|---|---|
| **GENERAL COMMENTS** | This paper examines the relationship between television viewing and a number of potential causal intermediaries in the relationship between television viewing and poor health outcomes. The paper is ambitious in scope by looking at a number of factors and study designs and it is nice to see all of the variables described in one place. In general, the search is not well described and the results were difficult to follow.<br>It's not clear what the distinction between "intervention study" and experimental study" is.<br>The reasons for exclusions of studies and additional search terms were not clear (i.e., adults, additional search terms related to the outcomes of interest like energy expenditure or mental health).<br><br>Suggest changing to “randomized clinical trials” and “crossover studies” or between and within-person studies.<br>Line 259 and 266: Should “in our study” be “in one study”<br>The time of day section is not clear. IF the question is “when people watch most TV” there’s consistent evidence about this from previous day recall and time use studies<br>The section headers in results could be more informative. Instead of “characteristics of sitting” or “(in) activity” it could be “relationship between television and characteristics of sitting.” The decision to split up by study design was also a bit confusing. It seems that focusing on the topics and describing within each topic whether they are experimental or cross-sectional would make it easier to follow.<br>There are randomized trials that have examined the impact of reductions in television viewing and found increases in physical activity (i.e., Otten https://pubmed.ncbi.nlm.nih.gov/20008695/ https://pubmed.ncbi.nlm.nih.gov/24094792/ )<br>The effect sizes are not well reported in the results and are not well described. |

| | |
|---|---|
| **REVIEWER** | Sabrina Ayoub-Charette<br>Toronto 3D Knowledge Synthesis and Clinical Trials Unit, Clinical Nutrition and Risk Factor Modification Centre, St. Michael’s Hospital, |

| | |
|---|---|
| | Toronto, Ontario, Canada, Nutritional Sciences |
| **REVIEW RETURNED** | 09-Feb-2021 |

| | |
|---|---|
| **GENERAL COMMENTS** | Overall great work! I think this is a very interesting systematic review, and very important research.<br><br>Major points:<br>- Why did you decide to exclude studies published in a language other than English? You can translate any study easily. Did you get reports that were non-English with your search?<br>- Why did you exclude studies of a particular health status? If this was the case, perhaps your question or title should be specific to "healthy" adults (although looking at your supplementary tables, I don't think you can make this claim). Further, you describe excluding, for example, studies which report only on overweight or obese adults, but in supplementary table 2, you include studies with ranges of BMI (from normal weight to obese), which weakens your rational for excluding those studies specific to overweight or obese adults. By including studies with a range of BMI, you still have available in your sample overweight/obese adults, but at a smaller number which may bias the results. I would include all studies, regardless of their health status, and perhaps elaborate in the result section how their health status may affect the results observed.<br>- Perhaps it would be a better option to include all studies which meet the eligibility criteria in the SRMA, regardless of their quality assessment, and do separate interpretations on the studies with low vs. high internal validity. Otherwise, I would change the question to be specific to "… in studies with high standard of evidence"<br>- Figure 2: The fact that you originally identified 15000 records tells me that your search was not specific enough. Too late to change this now, you've already gone through all of them! However, on this figure, could you please describe why you excluded 1) the 10,000 reports after screening (e.g. x reports excluded for not directly answering our question, y reports excluded for non-English language etc.), 2)the 303 full-text records and 3)the 12 studies identified through 'other' sources – what are these other sources?<br>- Supplementary files 2-4: for all study designs (except cross-sectional), please include the duration of follow-up for the study<br><br>Minor points:<br>- Line 181-183: I am confused, what does this mean? Could you describe this better?<br>- Line 187-189: what does this mean? What was defined as a robust sampling method?<br>- In the results>search results section, please describe how many included studies were of observational and experimental designs, and how many studies were used in the assessment of each outcome you considered (you have already described this below, but I would include it in the search results section, to give an overview of what is to come in the following sections).<br>- Can you please define how you classified your studies as either experimental or intervention studies? To me, these are the same type of study design.<br>- I like how you described your results and the subheadings used!<br>- Within each outcome/study design, could you please describe your risk assessment results in the context of the study results?<br>- Heading at line 251: I would rename this heading to something more informative. All other headings used in the result section were simply e.g. "sleep" or "physical (in)activity" but here you have "characteristic of sitting" – please make this consistent. |

| | |
|---|---|
| | - Summary section beginning at line 479: Very interesting results! I think another mechanism that could explain these results is the lack of mindfulness while eating and watching TV, as TV distracts the watcher, and they may eat more than if they were only focused on eating.<br>- Line 598: what do you mean by this? “research is needed to examine how or whether TV and eating co-occur in free-living contexts.” I think intuitively, it is clear that many people do eat and watch TV at the same time. Are you just saying that we need formal research to quantify this?<br>- I wonder if you were able to find any studies which link alcohol intake and smoking to TV time? I’ve never looked into it, but I imagine that low SEP is linked to TV time, and also to higher alcohol/smoking rates, and that this may also confound the association with cardiometabolic health.<br>- I would include a few strengths in your discussion to highlight all the great work you did! |

**VERSION 1 – AUTHOR RESPONSE**

| **Reviewer 1** | |
|---|---|
| This paper examines the relationship between television viewing and a number of potential causal intermediaries in the relationship between television viewing and poor health outcomes. The paper is ambitious in scope by looking at a number of factors and study designs and it is nice to see all of the variables described in one place.<br><br>In general, the search is not well described and the results were difficult to follow. | As described above, we have refined the scope of the paper to include only studies that measured the outcome(s) of interest while watching TV (or immediately following) in an attempt to make the results clearer and easier to follow. We have also elaborated on the rationale for the search strategy (lines 106-111) and provided the full strategy as a supplementary file (supplementary file 1). |
| It's not clear what the distinction between "intervention study" and experimental study" is. | We made this distinction on the basis that experimental studies took place in laboratory contexts while intervention studies took place in free-living contexts. We have now merged these into one ‘experimental’ category. |
| The reasons for exclusions of studies and additional search terms were not clear (i.e., adults, additional search terms related to the outcomes of interest like energy expenditure or mental health). | Within our original scope, we had aimed to capture all studies that measured TV time in relation to a variable that was relevant to cardiometabolic health. For this reason, we did not add search terms to specify populations or outcomes of interest. We have added an explanation to this effect in the manuscript (lines 106-111). |
| Suggest changing to “randomized clinical trials” and “crossover studies” or between and within-person studies. | We have clarified our terminology to distinguish between between- and within-person studies (e.g., lines 162, 190) |
| Line 259 and 266: Should “in our study” be “in one | This is no longer relevant in the new version as |

| | |
|---|---|
| study" | this study and these lines have been removed. |
| The time of day section is not clear. IF the question is "when people watch most TV" there's consistent evidence about this from previous day recall and time use studies | This is no longer relevant in the new version as these studies have been removed. |
| The section headers in results could be more informative. Instead of "characteristics of sitting" or "(in) activity" it could be "relationship between television and characteristics of sitting." | We have updated the headings to be clearer: 'food and drink consumption while watching TV', 'characteristics of sitting while watching TV', 'affect during TV viewing', and 'the impact of TV viewing on sleep'. |
| The decision to split up by study design was also a bit confusing. It seems that focusing on the topics and describing within each topic whether they are experimental or cross-sectional would make it easier to follow. | This suggestion (in combination with your first comment) prompted us to refine the scope to improve the clarity and flow of the results. Now, with a focus only on studies that measured the outcome while (or immediately following) watching TV, we feel the results are much more cohesive. |
| There are randomized trials that have examined the impact of reductions in television viewing and found increases in physical activity (i.e., Otten https://pubmed.ncbi.nlm.nih.gov/20008695/ https://pubmed.ncbi.nlm.nih.gov/24094792/ ) | Thank you for mentioning these. We were aware of these studies but had excluded them on the basis that the study populations are exclusively overweight/obese adults, an exclusion criterion of our review. |
| The effect sizes are not well reported in the results and are not well described. | These were reported in Supplementary File 1 (now Supplementary File 2) where we calculated them ourselves or in Supplementary Files 3-4 where the study reported it. |
| **Reviewer 2** | |
| Why did you decide to exclude studies published in a language other than English? You can translate any study easily. Did you get reports that were non-English with your search? | We set this criterion due to constrained resources for this review. However, we wish to clarify that, in the end, no studies were excluded on the basis of language alone. This has been clarified in Figure 1. |
| Why did you exclude studies of a particular health status? If this was the case, perhaps your question or title should be specific to "healthy" adults (although looking at your supplementary tables, I don't think you can make this claim). Further, you describe excluding, for example, studies which report only on overweight or obese adults, but in supplementary table 2, you include studies with ranges of BMI (from normal weight to obese), which weakens your rational for excluding those studies specific to overweight or obese adults. By including studies with a range of BMI, you still have available in your sample overweight/obese adults, but at a smaller number which may bias the | We wanted to keep the scope of the review as generalizable to the broader adult population as possible. This was especially important because the majority of the literature base (including studies we cited in the paper's introduction) that has shown the effects of TV time on cardiometabolic health to be larger than the effects of total sedentary time have been primarily based on general adult populations. For this reason, we included studies whose samples included a range of BMIs (reflective of the population) but excluded studies whose samples were only comprised of overweight/obese individuals. The same logic extended to other |

| | |
|---|---|
| results. I would include all studies, regardless of their health status, and perhaps elaborate in the result section how their health status may affect the results observed. | health conditions: we did not automatically exclude studies if they included *any* participants with a given health condition but did exclude studies who *only* recruited participants with a health condition. |
| Perhaps it would be a better option to include all studies which meet the eligibility criteria in the SRMA, regardless of their quality assessment, and do separate interpretations on the studies with low vs. high internal validity. Otherwise, I would change the question to be specific to "… in studies with high standard of evidence" | In the end, few studies (n=4) were deemed to be of poor quality and, in the case of this particular review, it was not even possible to extract data from these studies because the reporting was of such poor quality. We therefore cannot include them in the review. At the same time, we did not restrict the review to high-quality studies only (the vast majority were of 'fair quality', see supplementary files 3-4). |
| Figure 2: The fact that you originally identified 15000 records tells me that your search was not specific enough. Too late to change this now, you've already gone through all of them! However, on this figure, could you please describe why you excluded 1) the 10,000 reports after screening (e.g. x reports excluded for not directly answering our question, y reports excluded for non-English language etc.), 2)the 303 full-text records and 3)the 12 studies identified through 'other' sources – what are these other sources? | We agree the search was broad but felt it had to be because 1) authors often describe studies as being about sedentary time or sedentary behaviour when in reality they are only about TV (thus we had to use all possible terms for capturing this), and 2) we did not have *a priori* expectations about what outcome variables should be included. We have added a statement to clarify this in the methods (lines 106-111) and have elaborated on the reasons for exclusion in Figure 1. |
| Supplementary files 2-4: for all study designs (except cross-sectional), please include the duration of follow-up for the study | This is no longer relevant as the new version does not contain studies with a follow-up period. |
| Line 181-183: I am confused, what does this mean? Could you describe this better? | This is no longer relevant in the new version. |
| Line 187-189: what does this mean? What was defined as a robust sampling method? | This is no longer relevant in the new version. We have, however, added a clarification that by 'robust' we mean within-subjects study designs to be more robust than between-subjects, and larger samples to be more robust than smaller samples (lines 161-163). |
| In the results>search results section, please describe how many included studies were of observational and experimental designs, and how many studies were used in the assessment of each outcome you considered (you have already described this below, but I would include it in the search results section, to give an overview of what is to come in the following sections). | We have attempted to make this clearer (lines 187-196). |
| Can you please define how you classified your studies as either experimental or intervention studies? To me, these are the same type of study | As noted above in response to reviewer 1, we made this distinction on the basis that experimental studies took place in laboratory |

| | |
|---|---|
| design. | contexts while intervention studies took place in free-living contexts. We have now merged these into one 'experimental' category. |
| I like how you described your results and the subheadings used! | Thank you |
| Within each outcome/study design, could you please describe your risk assessment results in the context of the study results? | Wherever possible, and particularly where findings varied considerably between studies, we discussed factors that contributed to the study's quality to help give the reader a sense of what evidence was more reliable (e.g., lines 255-259, 279-282). |
| Heading at line 251: I would rename this heading to something more informative. All other headings used in the result section were simply e.g. "sleep" or "physical (in)activity" but here you have "characteristic of sitting" – please make this consistent. | We have updated the headings to be clearer: 'food and drink consumption while watching TV', 'characteristics of sitting while watching TV', 'affect during TV viewing', and 'the impact of TV viewing on sleep'. |
| Summary section beginning at line 479: Very interesting results! I think another mechanism that could explain these results is the lack of mindfulness while eating and watching TV, as TV distracts the watcher, and they may eat more than if they were only focused on eating. | We agree this is a possible explanation but, as a key focus of this review is trying to assess evidence for speculative mechanisms, we are trying to avoid speculation as much as possible. |
| Line 598: what do you mean by this? "research is needed to examine how or whether TV and eating co-occur in free-living contexts." I think intuitively, it is clear that many people do eat and watch TV at the same time. Are you just saying that we need formal research to quantify this? | Yes, that is what we mean. We have elaborated on this in the discussion to clarify (lines 372-378). |
| I wonder if you were able to find any studies which link alcohol intake and smoking to TV time? I've never looked into it, but I imagine that low SEP is linked to TV time, and also to higher alcohol/smoking rates, and that this may also confound the association with cardiometabolic health. | This is a great question. There was one study captured in the review that examined TV time and alcohol intake but it was of such poor quality that it was excluded from the original review (it was also not eligible for the updated review because it did not look at alcohol consumption while watching TV). We did not find any studies that specifically addressed smoking while watching TV. These are interesting gaps that should be addressed in the future. |
| I would include a few strengths in your discussion to highlight all the great work you did! | Thank you. We have added this (lines 465-467). |

**VERSION 2 – REVIEW**

| | |
|---|---|
| **REVIEWER** | Sabrina Ayoub-Charette<br>Toronto 3D Knowledge Synthesis and Clinical Trials Unit, Clinical |

| | |
|---|---|
| | Nutrition and Risk Factor Modification Centre, St. Michael's Hospital, Toronto, Ontario, Canada, Nutritional Sciences |
| **REVIEW RETURNED** | 06-Apr-2021 |

| | |
|---|---|
| **GENERAL COMMENTS** | The authors addressed my all my comments in the response to reviewers. I appreciate the new scope of the review, and think that the narrowing of the scope benefits the study. Great work overall.<br><br>Major comments:<br>- There is a new PRISMA 2020 checklist that was released on the 29th of March 2021 - Consider implementing this!<br>- You have justified your use of particular health statuses, but I disagree. I think by including studies with any health conditions, but excluding studies with only a certain health condition, you are contradicting yourself. You will have more power in your analysis if you include all health conditions, then do exploratory subgroup analyses by health status later. Further, this will increase your generalizability to not only healthy adults, but all health statuses. Alternatively, you could only include studies on healthy normal weight adults or healthy mixed weight adults.<br>- Please add line 161-173 to your figures legend as well to help the readers follow the figure as they look at it without having to refer back to the text.<br>- Line 105: you cite Supplementary File 1 for your search strategy, but this file only contains the search strategy for PubMed - did you use the same search strategy for Web of Science? Please describe how you searched this.<br>- Consider changing your objective and conclusions to be 'in studies of fair and high quality' since you excluded/could not use the data from poor quality studies.<br>- Further, in line 187-196, please add that you only included studies of fair/high quality as the poor quality data was unusable.<br>- Line 255-273: The language you are using to describe observational results is not appropriate. All observational results should be discussed as 'associated with' or 'linked to'. Language should not be causal e.g. line 262: "there was no association between meal energy intake or duration while watching TV vs. no screens" etc.<br><br>Minor comments:<br>- Line 29: I would just call this 'eligibility criteria'<br>- Please add a short introductory sentence to your abstract.<br>- Line 48: As a strength, you also performed a systematic search, which is the greatest protection against bias in a review.<br>- In line 47-55: you describe here 2 strengths and 1 limitation, but in your discussion you only described 1 strength, and numerous limitations. I believe these sections should be consistent.<br>- Line 86-89: I think this would read better: 'the purpose of this critical systematic review is to improve our understanding of how TV time is linked to cardiometabolic health among adults. We have critically evaluated the available literature that provides evidence regarding food consumption, characteristics of sitting, or affect while watching TV, or on the impact of watching TV on subsequent sleep. [short sentence about results found]".<br>- Line 339: 'this critical systematic review aimed..."<br>- Line 157-159: this is hard to read. consider changing this to "we classified effect sizes as negligible (SMD<0.2, OR<1.50 or correlation <0.10), small (SMD..., OR... or correlation...), medium (SMD..., OR... or correlation...), or large (SMD..., OR... or correlation...)based on standard conventions" |

| | |
|---|---|
| | - to add to the discussion about experimental/intervention studies: to distinguish between lab vs. free-living context, you could use the terms 'in-patient' (=lab setting) and 'out-patient' (=free-living) re: results section. |

**VERSION 2 – AUTHOR RESPONSE**

| **Reviewer comments** | **Author responses** |
|---|---|
| **Major comments** | |
| There is a new PRISMA 2020 checklist that was released on the 29th of March 2021 - Consider implementing this! | We will certainly use the new checklist in future studies but have not done so here as this review was submitted well before the new checklist was published. |
| You have justified your use of particular health statuses, but I disagree. I think by including studies with any health conditions, but excluding studies with only a certain health condition, you are contradicting yourself. You will have more power in your analysis if you include all health conditions, then do exploratory subgroup analyses by health status later. Further, this will increase your generalizability to not only healthy adults, but all health statuses. Alternatively, you could only include studies on healthy normal weight adults or healthy mixed weight adults. | Our aim was to review studies that did not select by health condition, so that the findings are relevant to general populations (which include people with health conditions). This approach has been widely used elsewhere (for example, Chaudhry et al., 2020 https://doi.org/10.1186/s12966-020-01020-8 ; Barnett et al., 2017 https://doi.org/10.1186/s12966-017-0558-z ; Chastin et al., 2015 https://doi.org/10.1186/s12966-015-0292-3 ; Mclaughlin et al., 2021 https://doi.org/10.2196/23180) and we would prefer to retain it here. |
| Please add line 161-173 to your figures legend as well to help the readers follow the figure as they look at it without having to refer back to the text. | We have added this to the figure legends. |
| Line 105: you cite Supplementary File 1 for your search strategy, but this file only contains the search strategy for PubMed - did you use the same search strategy for Web of Science? Please describe how you searched this. | In the previous round of revision, the editor requested that we provide the search strategy for one database as a supplementary file. |
| Consider changing your objective and conclusions to be 'in studies of fair and high quality' since you excluded/could not use the data from poor quality studies. | Thank you for this useful addition. We have added this (line 343). |
| Further, in line 187-196, please add that you only included studies of fair/high quality as the poor quality data was unusable. | We have also added this (lines 185-186). |
| Line 255-273: The language you are using to describe observational results is not appropriate. All observational results should be | We recognise the issue she raises and have double-checked the language to ensure that we do not imply causality on the basis of observational studies. We |

| | |
|---|---|
| discussed as 'associated with' or 'linked to'. Language should not be causal e.g. line 262: "there was no association between meal energy intake or duration while watching TV vs. no screens" etc. | are happy that we have simply described the comparisons made and do not feel that we suggest causality. |
| **Minor comments:** | |
| Line 29: I would just call this 'eligibility criteria' | We have changed this (line 29). |
| Please add a short introductory sentence to your abstract. | We have followed *BMJ Open*'s guidance for authors which requires the abstract to be structured and begin with the objective (https://bmjopen.bmj.com/pages/authors/). |
| Line 48: As a strength, you also performed a systematic search, which is the greatest protection against bias in a review. | Thank you - this has been added (line 53). |
| In line 47-55: you describe here 2 strengths and 1 limitation, but in your discussion you only described 1 strength, and numerous limitations. I believe these sections should be consistent. | The bullet points are in line with the journal's requirement for strengths/limitations of the study methods while the strengths/limitations in the discussion primarily refer to the limitations of the evidence base itself, thus these are not inconsistent. However, we have listed an additional strength in the discussion (lines 469-470). |
| Line 86-89: I think this would read better: 'the purpose of this critical systematic review is to improve our understanding of how TV time is linked to cardiometabolic health among adults. We have critically evaluated the available literature that provides evidence regarding food consumption, characteristics of sitting, or affect while watching TV, or on the impact of watching TV on subsequent sleep. [short sentence about results found]". | We have rephrased this (lines 88-90). |
| Line 339: 'this critical systematic review aimed..." | We have changed this (line 342) |
| Line 157-159: this is hard to read. consider changing this to "we classified effect sizes as negligible (SMD<0.2, OR<1.50 or correlation <0.10), small (SMD..., OR... or correlation...), medium (SMD..., OR... or correlation...), or large (SMD..., OR... or correlation...)based on standard conventions" | We have changed this (lines 158-162). |
| to add to the discussion about experimental/intervention studies: to distinguish between lab vs. free-living context, you could use the terms 'in-patient' (=lab setting) and 'out-patient' (=free-living) re: | Thank you for this suggestion. However, as the participants were not patients we feel use of this terminology may be misleading and also possibly imply that the studies were clinical in nature. |

| | |
|---|---|
| results section. | |

## VERSION 3 – REVIEW

| | |
|---|---|
| **REVIEWER** | Sabrina Ayoub-Charette<br>Toronto 3D Knowledge Synthesis and Clinical Trials Unit, Clinical Nutrition and Risk Factor Modification Centre, St. Michael's Hospital, Toronto, Ontario, Canada, Nutritional Sciences |
| **REVIEW RETURNED** | 14-Apr-2021 |

| | |
|---|---|
| **GENERAL COMMENTS** | I have read your response to my previous comments, and all is good. Although I don't agree with all your decisions, your paper is good how it is. |